

Supplementary Material

Inferring effective interactions in a multispecies oral biofilm model:
Bayesian calibration and a testable prediction

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S1 Prior boxes

Table S1 lists the uniform prior box of every interaction entry for each condition and stage. The initial box was set empirically to contain earlier posterior estimates and encodes no sign constraint. An entry was widened only where more than 20% of its posterior mass lay in the outer 5% of the range on one side, or where this fraction increased by more than 0.10 between stages with the same likelihood; the stage was then repeated. The last column gives $\ln V = \sum_k \ln(u_k - l_k)$, the log-volume of the box, which enters $\ln \hat{Z}$ and is the reason why $\ln \hat{Z}$ is not compared across conditions.

Table S1: Prior boxes per condition and stage; widened entries in bold. **[generated by make_paper_figures.py from the final runs]**
[table]

S2 Runs and convergence

Table S2 lists, for every condition and stage, the number of particles and mutation steps, the number of tempering stages, the acceptance rate, the moves per parameter, and the spread over the three seeds of the maximum log-likelihood and of the posterior medians (criteria (i)–(iv) of the main text).

Table S2: Runs of the final pipeline. **[generated from the final runs]**
[table]

S3 Bayes factor for the *Fn–Pg* interaction

Stage (i) was repeated with $a_{45} = 0$ for all four conditions (three seeds each). Table S3 reports $\ln \hat{Z}$ of both models, $\ln B = \ln \hat{Z}_{\text{full}} - \ln \hat{Z}_{a_{45}=0}$, and the width of the a_{45} box, which sets the Occam penalty.

Table S3: Log-evidence of the full and the reduced model. **[CS and CH reduced-model runs with 120 mutation steps pending]**
[table]

Dysbiotic Static. In DS, a_{45} and a_{33} trade off. With $a_{45} = 0$ the posterior moves almost entirely to the mode $a_{33} \approx -10$ (weight **[0.83–0.87]**), whereas the full model places most of its mass at $a_{33} \approx +1$ (weight of the mode $a_{33} < -5$: **[0.09–0.12]**). Table S4 gives a_{45} separately for each mode.

Table S4: DS: a_{45} by a_{33} mode (stage **[iv]**, three seeds).

Mode	Particles	a_{45} median [5%, 95%]	max log-likelihood
$a_{33} \geq -5$	[]	[+2.4 [+0.5, +4.4]]	[]
$a_{33} < -5$	[]	[+0.8 [−0.2, +1.4]]	[]

S4 Narrower final box (± 2 SD)

Stage (iv) was repeated with a box of ± 2 posterior standard deviations around the stage-(iii) mean. **[comparison of medians and 90% intervals with the ± 4 SD runs, if available before submission; otherwise remove this section and the reference in the main text]**

S5 Likelihood weights

Stage (i) for DH was repeated with $\lambda_{Pg} = \lambda_{late} = 1$ (1000 particles, 100 mutation steps, three seeds; all convergence criteria met). Table ?? compares it with the weighted run.

Table S5: DH, stage (i): weighted ($\lambda_{Pg} = 5$, $\lambda_{late} = 3$) and unweighted likelihood. Ranges over three seeds.

	a_{45} median [5%, 95%]	Pg D21/D15 (measured 2.74)	RMSE
Weighted	+2.9 to +3.2 [+0.5 to +1.1, +5.2 to +5.5]	2.4–2.9	0.058–0.065
Unweighted	+3.0 [−0.06 to +0.17, +5.6]	1.7–3.1	0.059–0.068